# Practical No. 6

## Predictive Analytics for Personalized Healthcare Recommendations

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### Aim

To develop a predictive analytics system that provides personalized healthcare recommendations using patient data.

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### Learning Outcomes

- Understand predictive analytics concepts
- Apply machine learning for healthcare data
- Generate personalized health recommendations

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### Dataset Link

☞ **Healthcare Real-Time Risk Dataset** – Kaggle (simulated health records):
https://www.kaggle.com/datasets/alekhyaabburi/healthcare (Kaggle)

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### Theory

Predictive analytics uses historical patient data to forecast health risks and recommend treatments.
Machine learning models help in early disease detection and personalized healthcare planning.

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### System Architecture

Patient Dataset → Data Preprocessing → ML Model
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Risk Prediction
↓
Healthcare Recommendations

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## Procedure

**Step 1:** Load patient dataset
**Step 2:** Preprocess data (cleaning & normalization)
**Step 3:** Train predictive model
**Step 4:** Generate recommendations based on predictions

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## Output

- Disease risk prediction
- Personalized healthcare suggestions

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## Result

A predictive healthcare recommendation system was successfully developed.

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## Applications

- Hospitals
- Health monitoring apps
- Preventive healthcare systems

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## Advantages

- Early disease detection
- Personalized treatment plans

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## Precautions

- Ensure patient data privacy
- Use balanced datasets

## Conclusion

The experiment shows how predictive analytics improves personalized healthcare recommendations.

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## Viva Voce Questions

1. What is predictive analytics?
2. Why is ML important in healthcare?

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